

ACToR: CodeWeaver evidence profile

This profile audits every empirical surface identified for ACToR and records the maximum scientifically defensible CodeWeaver comparison. Status: exact_micro_campaign_and_blocked_macro. The six-program absolute hidden oracle is executable and receives three leakage-safe CodeWeaver repetitions; the 57-program macro metric is relative cross-testing without released fixed outputs/tests.

Evidence boundary

Paper values and CodeWeaver measurements remain separate unless subject identity, revision, denominator, and fixed oracle are verified. Missing values are not converted to zeros.

Existing result

18/18 build; 14/18 pass all; 1362/1476 hidden cases passed
(results/crust-citation-complete-codeweaver-2026-08-20/data/actor-li).

Complete empirical-surface audit

key	surface	denominator	metrics	status
actor-li	Figures 2 and 4	6 micro + 57 BSD CLI utilities	hidden/differential-test pass by 7 configurations	exact_micro_codeweaver_campaign_and_reference_only_macro
actor-li	C2SaferRust augmentation	7 standalone executables	differential-test pass	exact_reference_public_artifact
actor-li	Stability and cost ablations	3 runs and iteration/test/seed configurations	mean, standard deviation, accuracy and cost	exact_reference_public_artifact

Inclusion and execution decision

paper_id	source_url	empirical_scope	crust_role	codeweaver_status	reason	existing_result
arXiv:2510.03879	https://arxiv.org/abs/2510.03879	63 command-line C utilities	citation only	exact_micro_campaign_and_blocked_macro	the six-program absolute hidden oracle is executable and receives three leakage-safe CodeWeaver repetitions; the 57-program macro metric is relative cross-testing without released fixed outputs/tests	results/crust-citation-complete-codeweaver-2026-08-20/data/actor-li

actor_li_reference

surface	scope	system	metric	value	uncertainty
micro evaluation	6 utilities / 3 runs	ACToR Claude Code Sonnet 4.5	hidden-test pass percent	97.0	SD 1.9 pp
micro evaluation	6 utilities	naive Claude Code Sonnet 4.5	hidden-test pass percent	89.2	
micro evaluation	6 utilities / 10 iterations	ACToR Claude Code Sonnet 4.5	hidden-test pass percent	98.2	
macro evaluation	57 BSD utilities	coverage baseline	relative pass percent	58.4	
macro evaluation	57 BSD utilities	ACToR	relative pass percent	95.1	
C2SaferRust augmentation	7 executables	C2SaferRust	pass percent	76.3	
C2SaferRust augmentation	7 executables	C2SaferRust + ACToR	pass percent	92.9	
cost	57 BSD utilities	coverage baseline	USD	808	
cost	57 BSD utilities	ACToR	USD	1634	

CodeWeaver six-program hidden-oracle result

subject	cells	build	pass_all	safe_pass_all	hidden_tests	test_rate_percent
csplit	3	3/3	2/3	2/3	209/210	99.52%
expr	3	3/3	2/3	2/3	200/300	66.67%
fmt	3	3/3	3/3	3/3	198/198	100.00%
join	3	3/3	2/3	2/3	250/252	99.21%
printf	3	3/3	2/3	2/3	238/249	95.58%
test	3	3/3	3/3	3/3	267/267	100.00%
ALL	18	18/18	14/18	14/18	1362/1476	92.28%

CodeWeaver execution telemetry

subject	elapsed_hours	aiu	premium_requests	output_tokens
csplit	4.17	4909.659	49	unavailable
expr	3.53	4231.305	61	unavailable
fmt	3.62	2542.231	53	unavailable
join	4.17	3105.143	59	unavailable
printf	4.17	2609.613	55	unavailable
test	3.36	2367.341	60	unavailable
ALL	23.02	19765.291	337	unavailable